

The top-weight generating function of $\Psi(e_2^b)$

An exposition of a closed form, and of the weight bound it implies

Abstract

Let Ψ be the linear map on symmetric polynomials in three variables that sends each Schur function s_μ to the corresponding *shifted* Schur function s_μ^* of Okounkov and Olshanski. Writing the image in the basis of elementary symmetric polynomials E_1, E_2, E_3 and grading that polynomial ring by $w(E_1) = w(E_2) = 1, w(E_3) = 2$, we describe a closed form for the exponential generating function of the top-weight components of $\Psi(e_2^b)$: it factors as $(1 + E_1 T)^{E_2/E_1 - 1} \cdot \exp(E_3 M(T))$ for an explicit $M(T)$. We explain the proof — an operator recursion for $\Psi(e_2^b)$, a projection to top weight, and a uniqueness argument for a functional-differential equation — and we give a careful account of the corollary that motivated the whole computation: $\Psi(e_2^b)$ has weight at most b . That bound asserts a total, exceptionless cancellation among the $\lfloor b/2 \rfloor$ highest weight layers of a sum of shifted Schur functions with strictly positive coefficients. All statements are labelled as proved or as verified only in a finite range.

1 Introduction

The object

Fix three variables. The ring $\Lambda_3 = \mathbb{Q}[e_1, e_2, e_3]$ of symmetric polynomials in u_1, u_2, u_3 carries two natural bases indexed by partitions μ with at most three parts: the Schur polynomials s_μ , and the *shifted Schur polynomials* s_μ^* of Okounkov–Olshanski. The second family is the “inhomogeneous” analogue of the first: each s_μ^* is a polynomial with $s_\mu^* = s_\mu + (\text{terms of lower degree})$, and it is characterised by a vanishing property at partition points. Both families are bases of the same ring, so there is a unique linear isomorphism

$$\Psi : \Lambda_3 \longrightarrow \Lambda_3, \quad \Psi(s_\mu) = s_\mu^*.$$

This map is the transition matrix between the two bases. It is *not* a ring homomorphism, and that is exactly what makes it interesting: one wants to know how it interacts with products, and in particular what it does to powers of a fixed symmetric function.

The simplest nontrivial product to iterate is e_2 . So one asks: what is $\Psi(e_2^b)$?

Why anyone should care

Two reasons, one internal and one external.

Internally, $\Psi(e_2^b)$ is a single, concrete, completely computable family that measures the failure of Ψ to be multiplicative. Because Ψ is degree-filtered rather than graded, $\Psi(e_2^b)$ is an inhomogeneous polynomial whose top-degree part is e_2^b and whose lower-degree part is a genuine invariant of the shifted theory. Understanding it completely — a closed generating function, a support description, a sign rule — is a model case for “how does the shifted basis distort a product?”

Externally, the family arises when one pushes shifted Schur functions through a one-parameter specialisation. Substitute $(u_1, u_2, u_3) = (t, y, c)$, where y, c are the two roots of a quadratic with $y + c = s$ and $yc = t$ for an auxiliary parameter s . Then

$$e_1 \mapsto t + s, \quad e_2 \mapsto t(s + 1), \quad e_3 \mapsto t^2,$$

so the t -degrees of the images of e_1, e_2, e_3 are 1, 1, 2. Consequently, if one assigns to a monomial $E_1^{x_1} E_2^{x_2} E_3^{x_3}$ the *weight* $x_1 + x_2 + 2x_3$, then the maximum weight occurring in a symmetric polynomial is an upper bound for the t -degree of its specialisation. Bounding the weight of $\Psi(e_2^b)$ therefore bounds the t -degree of a family of specialised shifted Schur sums — and this is the form in which the question originally arose. The reader who does not care about the specialisation may simply take the $(1, 1, 2)$ -weight as a grading given by fiat; everything below is intrinsic.

The main theorem in one paragraph

Write $\Psi_b := \Psi(e_2^b)$, expressed as a polynomial in the elementary symmetric polynomials E_1, E_2, E_3 , and let $\text{tops}[b]$ be the part of Ψ_b of top $(1, 1, 2)$ -weight, namely weight b . Then the exponential generating function $F(T) = \sum_{b \geq 0} \text{tops}[b] T^b / b!$ factors as a product of two elementary series: a binomial factor $(1 + E_1 T)^{E_2/E_1 - 1}$, which involves only E_1 and E_2 and whose coefficients are the products $\prod_{r=1}^n (E_2 - r E_1)$; and an exponential factor $\exp(E_3 M(T))$, which carries all the E_3 -dependence, where $M(T) = \sum_{n \geq 2} (-1)^{n-1} \frac{n^2-1}{n} E_1^{n-2} T^n$. The two factors are “orthogonal”: one knows nothing about E_3 , the other nothing about E_2 . Equivalently, F satisfies the first-order linear ODE $F'/F = (E_2 - E_1)/(1 + E_1 T) - E_3 T(3 + E_1 T)/(1 + E_1 T)^3$.

The corollary the reader should care most about, and which is proved along the way rather than deduced from the closed form, is that $\Psi(e_2^b)$ has weight at most b — a statement of massive cancellation, discussed at length in §5.

Provenance and honesty conventions

The results below are due to Rick (autonomous research agent); this document is an exposition, not a research announcement. Every assertion is labelled. “Theorem”, “Lemma”, “Proposition”, “Corollary” means *proved*, with the proof given or sketched here. “Observation” means *verified computationally in a stated finite range and not proved*; such statements are never used in any proof. Section 7 collects the status of everything.

2 Definitions, and two worked examples

2.1 Shifted Schur polynomials in three variables

For $k \geq 0$ write $[x]_k := x(x-1) \cdots (x-k+1)$ for the falling factorial, and set

$$V := V(u_1, u_2, u_3) = (u_1 - u_2)(u_1 - u_3)(u_2 - u_3).$$

Definition 2.1. For a partition $\mu = (\mu_1 \geq \mu_2 \geq \mu_3 \geq 0)$ put $k_j := \mu_j + 3 - j$, so $k_1 > k_2 > k_3 \geq 0$. The *shifted Schur polynomial* is

$$s_\mu^*(u_1, u_2, u_3) := \frac{\det [[u_i]_{k_j}]_{i,j=1}^3}{V(u_1, u_2, u_3)}.$$

Since $[x]_k = x^k + (\text{lower})$ is monic of degree k , the numerator is an antisymmetric polynomial, hence divisible by V , and s_μ^* is a symmetric polynomial. Replacing $[u_i]_{k_j}$ by $u_i^{k_j}$ gives the Weyl determinant formula for s_μ ; comparing leading terms gives $s_\mu^* = s_\mu + (\text{lower degree})$. These are the shifted Schur functions of Okounkov–Olshanski written in the variables $u_i = x_i + 3 - i$, in which they become honestly symmetric.

Definition 2.2. $\Psi : \Lambda_3 \rightarrow \Lambda_3$ is the linear map with $\Psi(s_\mu) = s_\mu^*$ for all μ with at most three parts.

Lemma 2.3 (Operator form of Ψ). *Let T be the linear operator on $\mathbb{Q}[u_1, u_2, u_3]$ defined on monomials by*

$$T(u_1^{a_1} u_2^{a_2} u_3^{a_3}) := [u_1]_{a_1} [u_2]_{a_2} [u_3]_{a_3}.$$

Then for every symmetric f one has $\Psi(f) = T(fV)/V$.

Proof. T acts one variable at a time and therefore commutes with the permutation action, so $T(fV)$ is antisymmetric and divisible by V ; thus $f \mapsto T(fV)/V$ is a well-defined linear map. On $f = s_\mu$ we have $s_\mu V = \det[u_i^{k_j}]$, and applying T to the expanded determinant replaces each $u_i^{k_j}$ by $[u_i]_{k_j}$, giving $T(s_\mu V) = \det[[u_i]_{k_j}] = s_\mu^* V$. Two linear maps agreeing on a basis are equal. $\square$

Lemma 2.3 is what makes the whole subject computable: it converts a statement about a change of basis into a statement about the single explicit operator T .

2.2 The target ring and the $(1, 1, 2)$ -weight

We keep two notations for the same objects, to separate input from output. On the input side e_1, e_2, e_3 denote the elementary symmetric polynomials in u_1, u_2, u_3 . On the output side we regard Λ_3 as the polynomial ring $\mathbb{Q}[E_1, E_2, E_3]$, $E_i := e_i(u_1, u_2, u_3)$, and we always write elements of the image of Ψ as polynomials in E_1, E_2, E_3 .

Definition 2.4. The $(1, 1, 2)$ -weight on $\mathbb{Q}[E_1, E_2, E_3]$ is $w(E_1^{x_1} E_2^{x_2} E_3^{x_3}) := x_1 + x_2 + 2x_3$, extended to a polynomial P by $w(P) := \max\{w(m) : m \text{ a monomial with nonzero coefficient in } P\}$, with $w(0) := -\infty$. Weight is additive on products of monomials, so $w(PQ) \leq w(P) + w(Q)$, and the weight- $\leq w$ subspaces form a ring filtration. We write $P|_{w=w}$ for the sum of the terms of P of weight exactly w .

Note that the weight is not the degree: $E_1^{x_1} E_2^{x_2} E_3^{x_3}$ has degree $x_1 + 2x_2 + 3x_3$ in the u_i , but weight $x_1 + x_2 + 2x_3$. The weight is the pullback of the t -degree along the specialisation described in the introduction.

Definition 2.5. $\Psi_b := \Psi(e_2^b) \in \mathbb{Q}[E_1, E_2, E_3]$ and $\text{tops}[b] := \Psi_b|_{w=b}$.

2.3 Worked example 1: $b = 1$ and $b = 2$ by hand

Example 2.6. $e_2 = s_{(1,1,0)}$, so $\Psi_1 = s_{(1,1,0)}^*$. Here $k = (3, 2, 0)$ and

$$s_{(1,1,0)}^* V = \det \begin{pmatrix} [u_1]_3 & [u_1]_2 & 1 \\ [u_2]_3 & [u_2]_2 & 1 \\ [u_3]_3 & [u_3]_2 & 1 \end{pmatrix},$$

which expands to $(E_2 - E_1 + 1)V$. Hence

$$\boxed{\Psi_1 = E_2 - E_1 + 1}$$

of weight 1. Its top-weight part is $\text{tops}[1] = E_2 - E_1$.

Example 2.7. By the Pieri/Kostka expansion $e_\lambda = \sum_\mu K_{\mu'\lambda} s_\mu$ with $\lambda = (2, 2)$,

$$e_2^2 = s_{(2,2,0)} + s_{(2,1,1)}, \quad \text{so} \quad \Psi_2 = s_{(2,2,0)}^* + s_{(2,1,1)}^*.$$

The second summand can be computed with no determinant at all. Indeed $s_{(2,1,1)} = e_3 \cdot s_{(1,0,0)}$, and Lemma 4.3 below gives $\Psi(e_3 f) = E_3 \cdot \Psi(f)(u - 1)$; since $s_{(1,0,0)}^* = E_1 - 3$ and E_1 becomes $E_1 - 3$ under $u \mapsto u - 1$, we get

$$s_{(2,1,1)}^* = E_3 (E_1 - 6).$$

Direct expansion of the other determinant gives $s_{(2,2,0)}^* = E_2^2 - 3E_1 E_2 + 2E_1^2 + 3E_3 + 5E_2 - 6E_1 + 4$, whence

$$\boxed{\Psi_2 = E_2^2 - 3E_1 E_2 + 2E_1^2 - 3E_3 + 5E_2 - 6E_1 + 4}$$

of weight 2, with $\text{tops}[2] = E_2^2 - 3E_1 E_2 + 2E_1^2 - 3E_3$.

Example 2.7 already contains the phenomenon this paper is about. The two summands $s_{(2,2,0)}^*$ and $s_{(2,1,1)}^*$ have weights 3 and 3: their weight-3 parts are $-E_1 E_3$ and $+E_1 E_3$. They cancel. The sum has weight 2, one less than either summand. We return to this in §5.

For orientation, here are the next two values (computed from Lemma 2.3):

$$\begin{aligned} \text{tops}[3] &= E_2^3 - 6E_1 E_2^2 + 11E_1^2 E_2 - 6E_1^3 - 9E_2 E_3 + 25E_1 E_3, \\ \text{tops}[4] &= E_2^4 - 10E_1 E_2^3 + 35E_1^2 E_2^2 - 50E_1^3 E_2 + 24E_1^4 \\ &\quad - 18E_2^2 E_3 + 118E_1 E_2 E_3 - 190E_1^2 E_3 + 27E_3^2. \end{aligned}$$

3 The main theorem

Work in $R := \mathbb{Q}[E_1, E_2, E_3]$ and in the formal power series ring $R[[T]]$.

Definition 3.1. Define $A(T), B(T) \in R[[T]]$ by

$$\begin{aligned} A(T) &:= \sum_{n \geq 0} \frac{A_n}{n!} T^n, \quad A_n := \prod_{r=1}^n (E_2 - r E_1), \\ B(T) &:= \exp(E_3 M(T)), \quad M(T) := \sum_{n \geq 2} (-1)^{n-1} \frac{n^2 - 1}{n} E_1^{n-2} T^n. \end{aligned}$$

The exponential is legitimate because M has no constant term; and $[T^m]B(T)$ involves E_3^k only for $k \leq m/2$, so all coefficients lie in R .

The suggestive notation for A is $A(T) = (1 + E_1 T)^{E_2/E_1 - 1}$: formally expanding the binomial series and distributing one factor of E_1 into each bracket produces exactly $A_n = \prod_{r=1}^n (E_2 - r E_1)$. We will not use the fractional exponent; it does not live in R , and Definition 3.1 is the honest version. Likewise $M(T)$ has the closed form $T/(E_1(1 + E_1 T)^2) - \log(1 + E_1 T)/E_1^2$, which is a convenient mnemonic but again involves division by E_1 .

Theorem 3.2 (Closed form for the top-weight generating function). *Let $F(T) := \sum_{b \geq 0} \text{tops}[b] T^b/b! \in R[[T]]$. Then*

$$F(T) = A(T) \cdot B(T).$$

Equivalently, in terms of coefficients,

$$\text{tops}[b] = \sum_{n+m=b} \binom{b}{n} A_n B_m, \quad B_m := m! [T^m] B(T). \quad (1)$$

Corollary 3.3 (ODE form). *F is the unique solution in $R[[T]]$ with $F(0) = 1$ of*

$$\frac{F'(T)}{F(T)} = \frac{E_2 - E_1}{1 + E_1 T} - \frac{E_3 T (3 + E_1 T)}{(1 + E_1 T)^3},$$

i.e. of $(1 + E_1 T)^3 F' = [(E_2 - E_1)(1 + E_1 T)^2 - E_3 T(3 + E_1 T)] F$. Extracting coefficients gives the three-term recursion

$$\begin{aligned} \text{tops}[b+1] &= (E_2 - (3b+1)E_1) \text{tops}[b] + b(2E_1 E_2 - (3b-1)E_1^2 - 3E_3) \text{tops}[b-1] \\ &\quad + b(b-1)(E_1^2 E_2 - (b-1)E_1^3 - E_1 E_3) \text{tops}[b-2]. \end{aligned}$$

Remark 3.4 (Weight homogeneity). Assign T the weight -1 . Then A_n has weight n (its top monomial is E_2^n), and each monomial $E_3^k E_1^{m-2k}$ of B_m has weight $2k + (m-2k) = m$. So A, B, F are all homogeneous of weight 0 and $\text{tops}[b]$ is homogeneous of weight exactly b — as it must be, being a top-weight component.

Remark 3.5 (Structural reading). The factorisation separates variables in a strong sense: A is free of E_3 , B is free of E_2 . So the whole E_3 -dependence of the top-weight data is an exponential of a single explicit series, and the whole E_2 -dependence is a shifted-factorial product. This is not just a compact formula; the two factors carry independent alternating signs which align perfectly, and that alignment is what makes the coefficient-level results of §6 possible.

3.1 Worked example 2: reading $\text{tops}[2]$ and $\text{tops}[4]$ off the formula

Example 3.6. From Definition 3.1: $A_0 = 1$, $A_1 = E_2 - E_1$, $A_2 = (E_2 - E_1)(E_2 - 2E_1)$. The coefficients of M are $\mu_2 = -\frac{3}{2}$, $\mu_3 = \frac{8}{3}$, $\mu_4 = -\frac{15}{4}$, so $B_0 = 1$, $B_1 = 0$, $B_2 = 2! \mu_2 E_3 = -3E_3$. By (1),

$$\text{tops}[2] = \binom{2}{0} A_0 B_2 + \binom{2}{1} A_1 B_1 + \binom{2}{2} A_2 B_0 = -3E_3 + 0 + E_2^2 - 3E_1 E_2 + 2E_1^2,$$

matching Example 2.7. *Check by hand.*

Example 3.7 (The pure E_3 corner). Take $b = 4$ and ask for the coefficient of E_3^2 . Only $n = 0$ contributes (A_n has no E_3 , and B_m needs $m \geq 2k = 4$), so the coefficient is $[E_3^2] B_4 = 4! [T^4] \frac{1}{2} M(T)^2 = \frac{4!}{2} \mu_2^2 = 12 \cdot \frac{9}{4} = 27$, using that the only way to write 4 as an ordered sum of two parts each ≥ 2 is $2 + 2$. And indeed $\text{tops}[4]$ above has $+27E_3^2$. The same computation for general even $b = 2\ell$ gives one composition $(2, \dots, 2)$ and

$$[E_3^{b/2}] \text{tops}[b] = \frac{b!}{\ell!} \left(\frac{\mu_2}{1} \right)^\ell = (-3)^{b/2} (b-1)!!,$$

e.g. $-3, 27, -405, 8505$ for $b = 2, 4, 6, 8$. *Check the $b = 6$ value against $\text{tops}[6]$ if you like; it is -405 .*

4 The shape of the argument

4.1 Strategy in prose

The naive approach — understand $\text{tops}[b]$ directly, or find a “top-weight collapse” lemma explaining the cancellations of §2.3 — does not work, because the cancellation is not visible term by term. The successful strategy inverts the order:

- (1) *Forget the weight. Find an exact recursion for the whole polynomial Ψ_b .* Because $\Psi = T(-V)/V$ and T satisfies a clean commutation identity with multiplication by u_i , multiplication by e_2 on the input side becomes an explicit second-order differential operator on the output side. Carrying this out and simplifying gives a three-term recursion for Ψ_b involving one extra ingredient: the substitution σ induced by the simultaneous shift $u_i \mapsto u_i - 1$.
- (2) *Read the weight bound off the recursion.* All coefficients in the recursion have small weight and σ preserves the weight filtration, so induction gives $w(\Psi_b) \leq b$ immediately. This is the corollary of §5, and it is obtained *before* the closed form, not from it.
- (3) *Project the recursion to top weight.* The projection replaces σ by its “leading part” σ_{top} , which is the simple substitution $E_2 \mapsto E_2 - 2E_1$.
- (4) *Convert the top-weight recursion into a functional-differential equation and solve it by uniqueness.* The equation is not an ODE: it relates F to $\tilde{F}(T) = F(T)|_{E_2 \mapsto E_2 - 2E_1}$. But it determines F uniquely from $F(0) = 1$, and one checks directly that $A \cdot B$ satisfies it.

4.2 Step 1: the operator identities

Let $D_i := u_i \partial / \partial u_i$ be the i -th Euler operator, $E := D_1 + D_2 + D_3$, and $e_2(D) := D_1 D_2 + D_1 D_3 + D_2 D_3$. Let σ denote the simultaneous shift $u_i \mapsto u_i - 1$; on generators,

$$\sigma(E_1) = E_1 - 3, \quad \sigma(E_2) = E_2 - 2E_1 + 3, \quad \sigma(E_3) = E_3 - E_2 + E_1 - 1. \quad (2)$$

Lemma 4.1 (Core umbral identity). *For every $h \in \mathbb{Q}[u_1, u_2, u_3]$ and every i : $T(u_i h) = u_i T(h) - T(D_i h)$.*

Proof. By linearity check on $h = u_i^a g$ with g free of u_i . Then $T(u_i h) = [u_i]_{a+1} T(g)$, while $u_i T(h) - T(D_i h) = (u_i - a)[u_i]_a T(g) = [u_i]_{a+1} T(g)$. $\square$

Lemma 4.2 (The e_2 identity for T). *For every $f \in \mathbb{Q}[u_1, u_2, u_3]$,*

$$T(e_2 f) = e_2 T(f) - \sum_{i=1}^3 u_i T((D_j + D_k) f) + T(e_2(D) f),$$

where in the i -th summand $\{i, j, k\} = \{1, 2, 3\}$.

Proof. Apply Lemma 4.1 twice to a single product, for $p \neq q$:

$$T(u_p u_q f) = u_p T(u_q f) - T(D_p(u_q f)) = u_p u_q T(f) - u_p T(D_q f) - u_q T(D_p f) + T(D_p D_q f),$$

using $D_p(u_q f) = u_q D_p f$ and one more application of Lemma 4.1 to $T(u_q D_p f)$. Summing over the three pairs $\{p, q\}$: the first terms give $e_2 T(f)$; the last give $T(e_2(D) f)$; and in the middle, for each fixed i the variable u_i appears paired with each of the other two indices, contributing $u_i T((D_j + D_k) f)$. $\square$

Lemma 4.3 (e_3 -shift). *$T(e_3 X) = e_3 \sigma(T(X))$ for all X ; consequently $\Psi(e_3 f) = E_3 \sigma(\Psi(f))$ for symmetric f .*

Proof. On a monomial, $[u_i]_{c+a} = [u_i]_c \cdot [u_i - c]_a$; taking $c = 1$ in each variable gives the first identity. For the second, $T(e_3 f V) = e_3 \sigma(\Psi(f) V) = e_3 \sigma(V) \sigma(\Psi(f))$, and $\sigma(V) = V$ because a simultaneous shift preserves differences. Divide by V . $\square$

Lemma 4.4 (e_1 -shift). *$\Psi(e_1 f) = (E_1 - 3)\Psi(f) - \Psi(Ef)$. In particular, if f is E -homogeneous of degree d then $\Psi(e_1 f) = (E_1 - d - 3)\Psi(f)$; e.g. $\Psi(e_1 e_2^b) = (E_1 - 2b - 3)\Psi_b$.*

Proof. By Lemma 4.1, $\Psi(e_1 f) = \sum_i (u_i T(fV) - T(D_i(fV)))/V$. The first sum is $E_1 \Psi(f)$. For the second, $D_i(fV) = D_i(f)V + f D_i(V)$ and $\sum_i D_i(V) = 3V$ since V is homogeneous of degree 3; so $\sum_i T(D_i(fV)) = T(E(f)V) + 3T(fV)$. $\square$

4.3 Step 2: the exact recursion for Ψ_b

Apply Lemma 4.2 to $f = e_2^b V$ and divide by V . Two elementary computations do all the work. Both use the same trick: a sum of rational functions over ordered pairs is grouped into unordered pairs, where the denominator cancels.

Lemma 4.5. $\sum_i u_i (D_j + D_k) (e_2^b V) = (2b+1) e_1 e_2^b V - b(e_1 e_2 - 3e_3) e_2^{b-1} V.$

Proof. Write $D_j + D_k = E - D_i$ and use $E(e_2^b V) = (2b+3)e_2^b V$. It remains to compute $\sum_i u_i D_i(e_2^b V) = b e_2^{b-1} V \sum_i u_i D_i(e_2) + e_2^b \sum_i u_i D_i(V)$. Now $\sum_i u_i D_i(e_2) = \sum_i u_i^2 (u_j + u_k) = e_1 e_2 - 3e_3$, and

$$\sum_i u_i D_i(V) = V \sum_{i \neq j} \frac{u_i^2}{u_i - u_j} = V \sum_{i < j} \frac{u_i^2 - u_j^2}{u_i - u_j} = V \sum_{i < j} (u_i + u_j) = 2e_1 V.$$

Combining gives the stated identity. $\square$

Lemma 4.6 (The critical simplification). $\sum_{\alpha \neq \beta} D_\alpha(e_2) D_\beta(V) = 3e_2 V$, and $\sum_\alpha D_\alpha(e_2) D_\alpha(V) = 3e_2 V$.

Proof. Since $E(e_2)E(V) = 2e_2 \cdot 3V = 6e_2 V$ and the two displayed sums add up to $E(e_2)E(V)$, the two statements are equivalent; we prove the second. With $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$, $D_\alpha(e_2) = u_\alpha(e_1 - u_\alpha)$ and $D_\alpha(V)/V = \sum_{\beta \neq \alpha} u_\alpha / (u_\alpha - u_\beta)$, so

$$\frac{1}{V} \sum_\alpha D_\alpha(e_2) D_\alpha(V) = \sum_{\alpha \neq \beta} \frac{u_\alpha^2 (e_1 - u_\alpha)}{u_\alpha - u_\beta}.$$

Group the ordered pairs (α, β) and (β, α) :

$$\frac{u_\alpha^2 (e_1 - u_\alpha) - u_\beta^2 (e_1 - u_\beta)}{u_\alpha - u_\beta} = e_1 (u_\alpha + u_\beta) - (u_\alpha^2 + u_\alpha u_\beta + u_\beta^2) = e_2,$$

the last equality by expanding $e_1 = u_\alpha + u_\beta + u_\gamma$. There are three unordered pairs, so the total is $3e_2$. $\square$

Lemma 4.7. $e_2(D)(e_2^b V)/V = (b+1)(b+2)e_2^b + b(b-1)e_1 e_3 e_2^{b-2}.$

Proof. Expand $D_\alpha D_\beta(e_2^b V)$ by the product rule and sum over $\alpha < \beta$. Four elementary evaluations are needed: $\sum_{\alpha < \beta} D_\alpha D_\beta(e_2) = e_2$; $\sum_{\alpha < \beta} D_\alpha(e_2) D_\beta(e_2) = e_2^2 + e_1 e_3$; $\sum_{\alpha < \beta} D_\alpha D_\beta(V) = 2V$; and $\sum_{\alpha \neq \beta} D_\alpha(e_2) D_\beta(V) = 3e_2 V$ (Lemma 4.6). The first three are direct expansions. Collecting the coefficient of e_2^b gives $b(b-1) + b + 3b + 2 = (b+1)(b+2)$, and the only other surviving term is $b(b-1)e_1 e_3 e_2^{b-2}$ from the $e_1 e_3$ part of the second evaluation. $\square$

Lemma 4.6 is the load-bearing step of the whole proof: a priori the sum $\sum_{\alpha \neq \beta} D_\alpha(e_2) D_\beta(V)$ is an unwieldy rational function, and its collapse to the scalar multiple $3e_2$ of V is what turns Lemma 4.7 into a clean quadratic polynomial in b .

Proposition 4.8 (Exact Ψ -recursion). *For all $b \geq 0$, with $\Psi_{-1} = \Psi_{-2} = 0$ and $\Psi_0 = 1$,*

$$\Psi_{b+1} = [E_2 - (b+1)E_1 + (b+1)^2] \Psi_b - 3b E_3 \sigma(\Psi_{b-1}) - b(b-1)(E_1 - 2b - 2) E_3 \sigma(\Psi_{b-2}). \quad (3)$$

Proof sketch. Write $\Psi_{b+1} = P_1 - P_2 + P_3$ for the three terms of Lemma 4.2 applied to $f = e_2^b V$ and divided by V ; so $P_1 = E_2 \Psi_b$ and $P_3 = \Psi(e_2(D)(e_2^b V)/V)$. Applying Lemma 4.1 once more inside P_2 produces $P_2 = 2P_3 + T(\sum_i u_i (D_j + D_k)(e_2^b V))/V$, because each unordered pair of indices occurs twice. Lemma 4.5 evaluates the remaining term, and Lemmas 4.4 and 4.3 convert it into $(b+1)(E_1 - 2b - 3)\Psi_b + 3b E_3 \sigma(\Psi_{b-1})$. Lemma 4.7 together with Lemmas 4.4, 4.3 gives $P_3 = (b+1)(b+2)\Psi_b + b(b-1)(E_1 - 2b - 2)E_3 \sigma(\Psi_{b-2})$. Assembling, $\Psi_{b+1} = P_1 - P_3 - [(b+1)(E_1 - 2b - 3)\Psi_b + 3b E_3 \sigma(\Psi_{b-1})]$, and $E_2 - (b+1)(E_1 - 2b - 3) - (b+1)(b+2) = E_2 - (b+1)E_1 + (b+1)^2$. $\square$

4.4 Step 3: projection to top weight

Lemma 4.9. σ preserves the weight filtration, and there is a unique ring endomorphism σ_{top} of $\mathbb{Q}[E_1, E_2, E_3]$ such that $\sigma(P)|_{w=w} = \sigma_{\text{top}}(P|_{w=w})$ whenever $w = w(P)$; explicitly

$$\sigma_{\text{top}}(E_1) = E_1, \quad \sigma_{\text{top}}(E_2) = E_2 - 2E_1, \quad \sigma_{\text{top}}(E_3) = E_3.$$

Proof. By (2), $\sigma(E_i)$ has weight $\leq w(E_i)$, and its top-weight part is as displayed; the filtration statement follows since σ is a ring map. Weight is additive on products of monomials, so taking top-weight parts is multiplicative on the associated graded, which makes σ_{top} a ring endomorphism. $\square$

Given the weight bound $w(\Psi_b) \leq b$ (Theorem 5.1 below, proved independently in §5), projecting (3) onto weight $b+1$ kills the constant $(b+1)^2$ and the constant $-2b-2$, leaving:

Proposition 4.10 (Top-weight recursion). *For all $b \geq 0$, with $\text{tops}[0] = 1$ and $\text{tops}[-1] = \text{tops}[-2] = 0$,*

$$\text{tops}[b+1] = (E_2 - (b+1)E_1)\text{tops}[b] - 3bE_3\sigma_{\text{top}}(\text{tops}[b-1]) - b(b-1)E_1E_3\sigma_{\text{top}}(\text{tops}[b-2]).$$

4.5 Step 4: the shift equation, and uniqueness

Write $\tilde{P} := \sigma_{\text{top}}(P) = P|_{E_2 \mapsto E_2 - 2E_1}$, extended coefficientwise to $R[[T]]$.

Proposition 4.11 (Shift equation). *$F(T)$ is the unique element of $R[[T]]$ with $F(0) = 1$ satisfying*

$$(1 + E_1T)F'(T) = (E_2 - E_1)F(T) - E_3T(3 + E_1T)\tilde{F}(T). \quad (4)$$

Proof. Multiply the recursion of Proposition 4.10 by $T^b/b!$ and sum over $b \geq 0$. The four sums are F' , $E_2F - E_1(TF)'$, $3E_3T\tilde{F}$ and $E_1E_3T^2\tilde{F}$ respectively; rearranging gives (4). Uniqueness: extracting $[T^b/b!]$ from (4) returns exactly the recursion of Proposition 4.10, which determines $\text{tops}[b+1]$ from $\text{tops}[b]$, $\text{tops}[b-1]$, $\text{tops}[b-2]$ and their (well-defined, polynomial) images under σ_{top} . $\square$

Note that (4) is not an ordinary differential equation: the substitution $E_2 \mapsto E_2 - 2E_1$ is a genuine extra operation, coupling F to a different function. This is why the “natural” equation is (4) rather than the ODE of Corollary 3.3: the latter is obtained by eliminating $\tilde{F}$, which costs a factor $(1 + E_1T)^{-2}$ and hence produces the cube in the denominator.

Proof of Theorem 3.2. Set $G := A \cdot B$; we verify (4) for G , with $G(0) = 1$.

(i) *A and B satisfy first-order equations.* From $A_{n+1} = (E_2 - (n+1)E_1)A_n$ one gets, comparing $[T^n/n!]$ of both sides, $(1 + E_1T)A' = (E_2 - E_1)A$. For B : the coefficient of T^n in $-T(3 + E_1T)(1 + E_1T)^{-3}$ is $-(-1)^{n-1}n(n+2)E_1^{n-1}$, which equals $(n+1)\mu_{n+1}E_1^{n-1} = [T^n]M'(T)$ with $\mu_n = (-1)^{n-1}(n^2-1)/n$; hence $(1 + E_1T)^3M' = -T(3 + E_1T)$, and $B'/B = E_3M'$ gives $(1 + E_1T)^3B' = -E_3T(3 + E_1T)B$.

(ii) *The substitution.* B is free of E_2 , so $\tilde{B} = B$. And $\tilde{A}_n = \prod_{r=1}^n (E_2 - (r+2)E_1)$, so $\tilde{A}(T) = A(T)(1 + E_1T)^{-2}$ in the sense that $(1 + E_1T)^2\tilde{A} = A$ (both sides satisfy the same first-order recursion with the same initial value; or expand the binomial series directly). Hence $(1 + E_1T)^2\tilde{G} = G$.

(iii) *Verification.* By (i), $G'/G = A'/A + B'/B = (E_2 - E_1)/(1 + E_1T) - E_3T(3 + E_1T)/(1 + E_1T)^3$. Multiplying by $(1 + E_1T)G$ and using (ii),

$$(1 + E_1T)G' = (E_2 - E_1)G - \frac{E_3T(3 + E_1T)}{(1 + E_1T)^2}G = (E_2 - E_1)G - E_3T(3 + E_1T)\tilde{G},$$

which is (4). By the uniqueness in Proposition 4.11, $F = G$. $\square$

5 The weight-bound corollary

5.1 Statement and proof

Theorem 5.1 (Weight bound). *For every $b \geq 0$, $w(\Psi(e_2^b)) \leq b$; in fact $w(\Psi(e_2^b)) = b$.*

Proof. Induction on b using the exact recursion (3), with $w(\Psi_0) = 0$. Assume $w(\Psi_j) \leq j$ for $j \leq b$. In (3):

- $E_2 - (b+1)E_1 + (b+1)^2$ has weight ≤ 1 , so the first term has weight $\leq b+1$;
- σ preserves the weight filtration (Lemma 4.9), so $E_3\sigma(\Psi_{b-1})$ has weight $\leq 2 + (b-1) = b+1$;
- likewise $(E_1 - 2b - 2)E_3\sigma(\Psi_{b-2})$ has weight $\leq 1 + 2 + (b-2) = b+1$.

Hence $w(\Psi_{b+1}) \leq b+1$. Sharpness: $\Psi(f) = f + (\text{lower degree})$ for homogeneous f , so the monomial E_2^b occurs in Ψ_b with coefficient 1, and it has weight b . $\square$

The proof is three lines, but note what it depends on: the exact recursion (3), whose derivation occupies all of §4. In particular the weight bound does *not* depend on Theorem 3.2; the logical order is recursion $\Rightarrow$ weight bound $\Rightarrow$ top-weight recursion $\Rightarrow$ closed form.

5.2 What the bound actually rules out

To appreciate the statement one must know how large the weight of a single shifted Schur polynomial can be. Expand e_2^b in the Schur basis using Kostka numbers,

$$e_2^b = \sum_{\substack{|\mu|=2b \\ \ell(\mu) \leq 3}} K_{\mu', (2^b)} s_\mu, \quad \text{so} \quad \Psi_b = \sum_{\mu} K_{\mu', (2^b)} s_\mu^*.$$

All the coefficients $K_{\mu', (2^b)}$ are *strictly positive integers*. So no cancellation can come from the outer coefficients; whatever cancellation occurs must happen inside the E -basis expansions of the individual s_μ^* .

Observation 5.2 (verified for $b \leq 5$; not used below). For μ in the support above, $w(s_\mu^*) = d_\mu := \mu_1 + \lfloor (|\mu| - \mu_1)/2 \rfloor$. (The companion statement that the largest d_λ over the Schur support of s_μ^* equals d_μ is a theorem; the E -basis weight version stated here has been checked by direct computation for all μ occurring for $b \leq 5$ and is quoted only for illustration.)

Taking $\mu = (b, b, 0)$ gives $d_\mu = b + \lfloor b/2 \rfloor = \lfloor 3b/2 \rfloor$. So the summands have weight as large as $\lfloor 3b/2 \rfloor$, while the sum has weight b .

The content of Theorem 5.1 is that every one of the weight layers $b+1, b+2, \dots, \lfloor 3b/2 \rfloor$ of $\sum_{\mu} K_{\mu', (2^b)} s_\mu^$ vanishes identically — not one coefficient survives, for any b — even though every outer coefficient in the sum is positive and every individual summand contributes to those layers.*

Quantitatively: the number of monomials of weight exactly w is $\#\{(x_1, x_2, x_3) \in \mathbb{Z}_{\geq 0}^3 : x_1 + x_2 + 2x_3 = w\} = \lfloor (w+2)^2/4 \rfloor$, so the number of coefficients forced to vanish is $\sum_{w=b+1}^{\lfloor 3b/2 \rfloor} \lfloor (w+2)^2/4 \rfloor \sim \frac{19}{96}b^3$, against $\sim \frac{1}{12}b^3$ coefficients that survive. Asymptotically more than twice as many coefficients are annihilated as are kept:

b	layers killed	coefficients annihilated	coefficients surviving
2	3	6	7
4	5–6	28	22
6	7–9	75	50
10	11–15	283	161
20	21–30	1910	946

Example 5.3 (The smallest instance, in full). $b = 2$, where $\Psi_2 = s_{(2,2,0)}^* + s_{(2,1,1)}^*$ (Example 2.7). In the E -basis:

$$\begin{aligned} s_{(2,2,0)}^* &= \underbrace{-E_1E_3}_{w=3} + \underbrace{(E_2^2 - 3E_1E_2 + 2E_1^2 + 3E_3)}_{w=2} + \underbrace{(5E_2 - 6E_1)}_{w=1} + \underbrace{4}_{w=0}, \\ s_{(2,1,1)}^* &= \underbrace{+E_1E_3}_{w=3} + \underbrace{-6E_3}_{w=2}. \end{aligned}$$

Both summands have weight $3 > b = 2$. Their weight-3 parts are negatives of each other and cancel exactly, and the weight-2 parts add to $E_2^2 - 3E_1E_2 + 2E_1^2 - 3E_3 = \text{tops}[2]$. This is the whole phenomenon in miniature: not a cancellation of like terms with opposite outer signs, but a conspiracy between the top-weight symbols of two structurally different shifted Schur polynomials.

5.3 Why the bound is the interesting part

Three reasons.

(a) *It is a rigidity statement, and rigidity statements are the ones that are hard to see.* Nothing in the definition of s_μ^* suggests that a K -weighted sum of them should lose $\lfloor b/2 \rfloor$ weight layers. The individual summands do not lose them; the bound is a property of the particular positive combination determined by e_2^b .

(b) *It is exactly the statement needed downstream.* Under the specialisation $(u_1, u_2, u_3) \mapsto (t, y, c)$ with $y + c = s$, $yc = t$ described in the introduction, weight becomes an upper bound for t -degree; hence Theorem 5.1 yields $\deg_t \Psi(e_2^b)|_{\text{spec}} \leq b$, a bound that had previously been verified only numerically for $b \leq 10$.

(c) *It upgrades from one parameter to three.* The single-parameter family e_2^b controls all e -monomials, because Ψ interacts simply with e_1 and with e_3 :

Lemma 5.4. *For symmetric g homogeneous of u -degree d , and $a, c \geq 0$:*

$$\Psi(e_1^a g) = [E_1 - d - 3]_a \Psi(g), \quad \Psi(g e_3^c) = \Psi(g)|_{u \mapsto u-c} \cdot \Psi(e_3^c), \quad \Psi(e_3^c) = \prod_{j=0}^{c-1} (E_3 - jE_2 + j^2 E_1 - j^3).$$

Proof. The first is Lemma 4.4 iterated. For the second and third, e_3^c is the monomial $u^{(c,c,c)}$ and $[u_i]_{c+a} = [u_i]_c [u_i - c]_a$, so $T(e_3^c h) = (\prod_i [u_i]_c) \cdot (T(h)|_{u \mapsto u-c})$; take $h = gV$ and use $V|_{u \mapsto u-c} = V$. Finally $\prod_{i=0}^{c-1} \prod_i (u_i - j) = \prod_{j=0}^{c-1} (E_3 - jE_2 + j^2 E_1 - j^3)$. $\square$

Corollary 5.5. *For all $a_1, a_2, a_3 \geq 0$, $w(\Psi(e_1^{a_1} e_2^{a_2} e_3^{a_3})) \leq a_1 + a_2 + 2a_3 = w(E_1^{a_1} E_2^{a_2} E_3^{a_3})$.*

Proof. The substitution $u \mapsto u - c$ preserves weight, since by (2) (with 1 replaced by c) it sends each E_i to a polynomial of weight $w(E_i)$. By Lemma 5.4,

$$\Psi(e_1^{a_1} e_2^{a_2} e_3^{a_3}) = [E_1 - 3a_3 - 2a_2 - 3]_{a_1} \cdot (\Psi_{a_2}|_{u \mapsto u-a_3}) \cdot \Psi(e_3^{a_3}),$$

whose three factors have weight $\leq a_1$, $\leq a_2$ (Theorem 5.1) and $= 2a_3$ respectively. $\square$

So the whole three-parameter statement — that Ψ does not increase $(1, 1, 2)$ -weight on e -monomials — reduces to the single family treated here, and is now a theorem.

6 What this closes, and what remains open

6.1 Closed

- **The closed form** (Theorem 3.2) and its ODE (Corollary 3.3).
- **The weight bound** $w(\Psi(e_2^b)) \leq b$ for all b (Theorem 5.1), previously known only for $b \leq 10$ by computation, and its three-parameter consequence (Corollary 5.5).
- **The complete structure of the top-weight component.** The factorisation $F = A \cdot B$ makes every coefficient of $\text{tops}[b]$ explicit through (1), and the two factors carry aligned signs. This yields, as separately proved theorems that we state here without proof: the support of $\text{tops}[b]$ is *all* of $\{(x_1, x_2, x_3) : x_1 + x_2 + 2x_3 = b\}$, of size $\lfloor (b+2)^2/4 \rfloor$ (no coefficient ever vanishes); and every coefficient has sign $(-1)^{x_1+x_3}$. The sign rule in fact holds for all of Ψ_b , not just its top-weight part, by a separate argument that conjugates the recursion (3) by the involution $E_1 \mapsto -E_1$, $E_2 \mapsto E_2$, $E_3 \mapsto -E_3$ and proves nonnegativity of the conjugated family.
- **The E_3 -free face.** Setting $E_3 = 0$ in the conjugated form of (3) kills every correction term, giving the product formula $\Psi_b|_{E_3=0} = \pm \prod_{k=1}^b (E_2 + kE_1 + k^2)$ up to the sign involution — an identity that simultaneously recovers all previously known “corner” formulas, e.g. $[E_1^b] \Psi_b = (-1)^{bb!}$ and $[E_2^b] \Psi_b = 1$.

6.2 Open

- **Other e_r .** Nothing here is specific to shifted Schur functions in an essential way, but a great deal is specific to $r = 2$: the argument needs the analogue of Lemma 4.6, i.e. the collapse of $\sum_{\alpha \neq \beta} D_\alpha(e_r) D_\beta(V)$ to a polynomial multiple of V . Whether that happens for $r \neq 2$ is not known.

- **More variables.** Everything above is written for three variables, where $\Lambda_3 = \mathbb{Q}[E_1, E_2, E_3]$ and V has degree 3. The shape of the argument should survive, but the weight $(1, 1, 2)$ and the constants $3, 2b + 3, (b + 1)(b + 2)$ are all specific to $n = 3$.
- **A combinatorial model.** The coefficients of Ψ_b , after the sign involution, are positive integers, so one expects them to count something. On the E_3 -free face the product formula $\prod_k (E_2 + kE_1 + k^2)$ gives an immediate model (three-colour each element of $\{1, \dots, b\}$). Beyond that face no such model is known, and a natural guess — set partitions with singletons weighted by $E_2 + kE_1 + k^2$ and pairs by an E_3 -carrying weight — is *refuted*: for $b = 3$ it forces a negative pair weight. Finding the right structure is open.
- **Non-recursive coefficient formulas below top weight.** Explicit closed forms are known for the top-weight component (Theorem 3.2) and for the E_3 -free face; the intermediate slices $[E_3^k]\Psi_b$, $k \geq 1$, are currently only given by sequential recursions.

7 Status of every claim

Statement	Status
Lemmas 2.3, 4.1, 4.2, 4.3, 4.4	proved (full proofs above)
Lemmas 4.5, 4.6, 4.7, 4.9, 5.4	proved (full proofs above)
Proposition 4.8 (exact recursion)	proved (sketch above; verified symbolically $b \leq 6$)
Proposition 4.10 (top-weight recursion)	proved; verified symbolically $b \leq 7$
Proposition 4.11 (shift equation, uniqueness)	proved
Theorem 3.2 ($F = AB$)	proved; verified symbolically $b \leq 6$
Corollary 3.3 (ODE and 3-term recursion)	proved; verified symbolically $b \leq 8$
Theorem 5.1 (weight bound)	proved
Corollary 5.5 (all e -monomials)	proved
Example 3.7 ($[E_3^{b/2}] = (-3)^{b/2}(b - 1)!!$)	proved (from Theorem 3.2)
Observation 5.2 ($w(s_\mu^*) = d_\mu$)	verified for $b \leq 5$ only; not proved
Support/sign statements in §6	proved elsewhere; quoted, not proved here
Existence of a positive combinatorial model	open; one natural guess refuted

Two remarks on the sources of this exposition. First, the identity of Lemma 4.2 and the collapse of Lemma 4.6 were originally recorded with computational verification in place of a complete argument; the proofs given above are complete and were checked symbolically. Second, the closed form for $A(T)$ is usually written as $(1 + E_1 T)^{E_2/E_1 - 1}$, which is not an element of $\mathbb{Q}[E_1, E_2, E_3][[T]]$; we have replaced that notation by the coefficient definition $A_n = \prod_{r=1}^n (E_2 - rE_1)$ and the first-order equation $(1 + E_1 T)A' = (E_2 - E_1)A$, which say the same thing without leaving the polynomial ring.